8 Appendices

8.1 Data Cleaning

- Missing or corrupted information was cleaned using an ID matching process between each hazard's claim frequency and severity data.
- Negative values were transformed using its absolute value to ensure compliance with the ranges provided in the Project Data Dictionary.
- Extreme outlier values were handled in accordance with the specified values in the data dictionary, either by bringing values to the closer prescribed range limit or by removing the data entirely.
- Exceptions were made to datasets with outlier values if more than 5% of the data were outside the given range. In these cases, all data would be kept unchanged.
- Data which remained incomplete was removed from the dataset, if and only if it accounted for less than 5% of the given dataset. This was the case for all hazard areas.

8.2 Exploratory Data Analysis

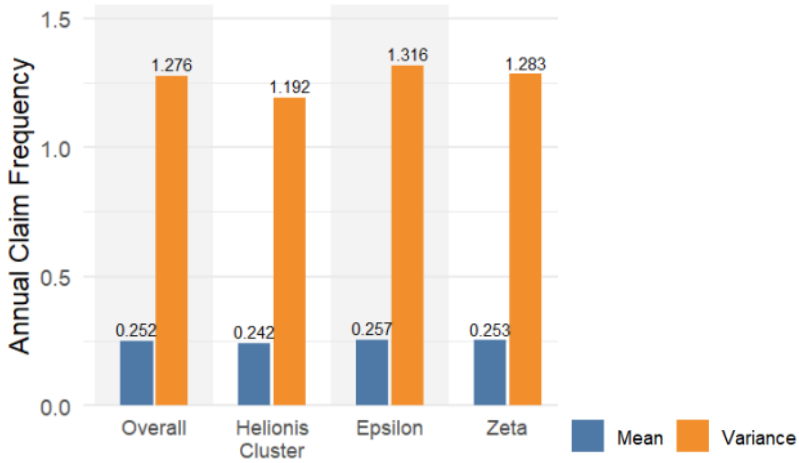
8.2.1 Business Interruption: Claim Frequency Behaviour and Simulation Process

	solar_system	n	max_amt	p99_amt	p999_amt	thr_iqr3_amt	n_above_iqr3
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	Epsilon	4094	134163670	3.72e7	8.89e7	514836464.	0
2	Helionis Cluster	1928	142583386	4.15e7	1.16e8	907245759.	0
3	Zeta	4033	123404314	3.03e7	6.06e7	544808612.	0

No data in claim amount was removed, since no claim amount of single claim exceeded the outlier boundary (Q3+3IQR)

Solar System	Policy amount	Exposure
Helionis Cluster	19950	9970
Epsilon	39943	19899
Zeta	40030	20022

A single policy had an average exposure of 0.5 in 2174. In both long-term and short-term simulations, the general settings included maintaining the same insurance capacity listed above.



The annual frequency of claims was at about the same level of about 0.25 across three systems. Meanwhile, the variances were much greater than their corresponding means, suggesting the incapability of fitting Poisson model.

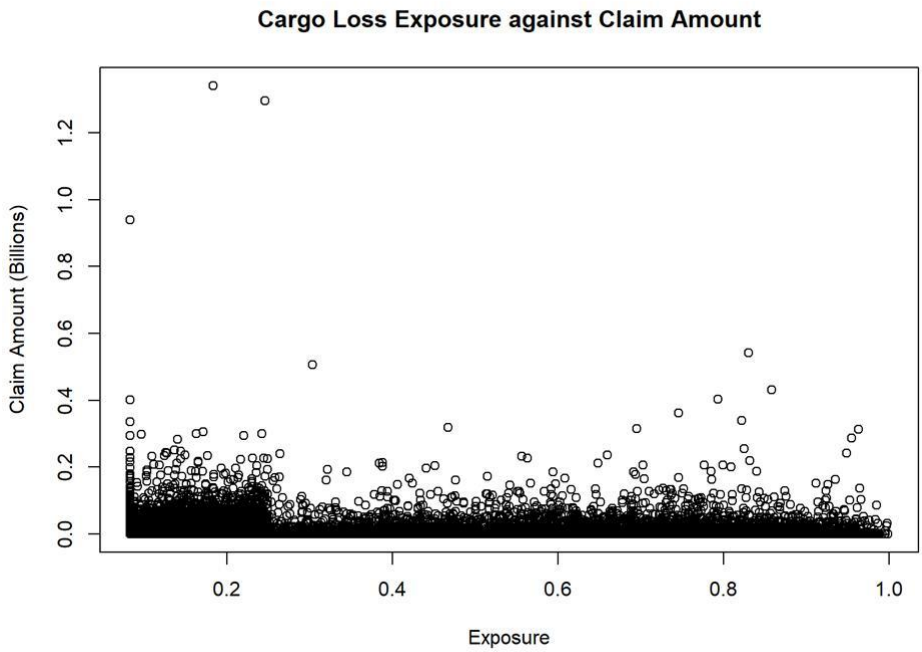
Short-term Expected Loss, Value at Risk (VaR) and TVaR were obtained by 10k times of Monte Carlo simulation.

Process:

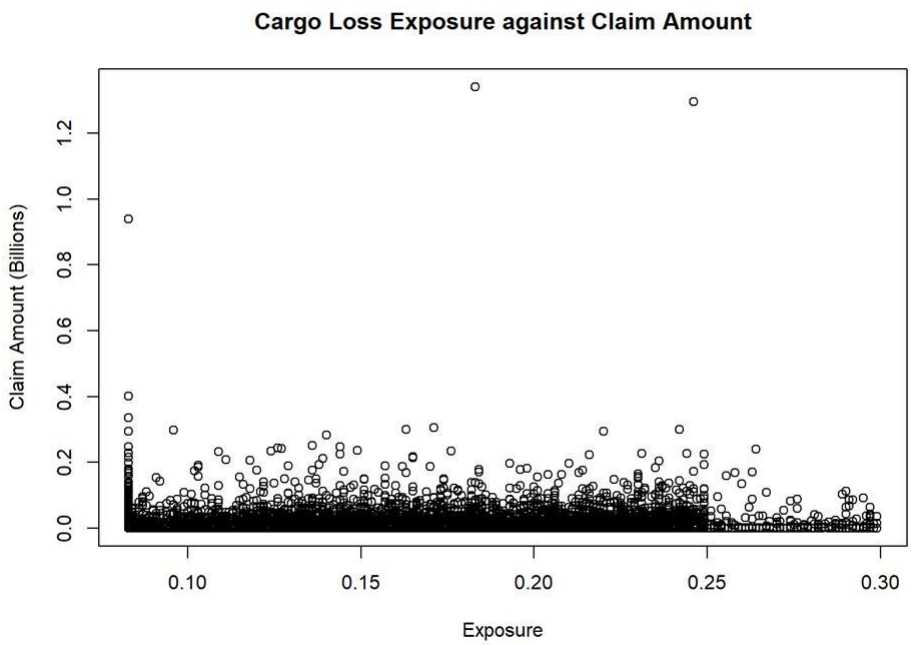
1. Obtain the annual frequency parameter of each solar system by fitting ZINB distribution
2. Shuffle the exposure of the policy
3. Obtain the exposure period frequency parameter (annual rate × exposure) of single policy (also ZINB distribution)
4. Generate random number for claim count for one policy
5. Generate random number for each claim

8.2.2 Cargo Loss: Low Exposure Policyholder Behaviour

Comparison of Exposure against Claim Amounts



It can be observed that there was an inflated number of claims for lower exposures compared to the rest of the exposures.



Upon closer inspection, it is clear that policyholders are buying the insurance product and making claims anticipating short-term losses. As a result, a waiting period of 3 months (0.25 of 1 year) was implemented.

8.2.3 Equipment Failure: Outlier Identification and Distribution Treatment

Outlier Statistic:

```
========================= freq Outlier Statistics =========================
       Variable Defined Range  Actual Min  Actual Max  Out of Bounds (Count) Out of Bounds (%)
  equipment_age       [3, 25]        3.00      327.36                    269             0.28%
maintenance_int   [100, 2000]      100.01   31,866.24                    152             0.16%
      usage_int      [0, 100]        0.00      467.89                     94             0.10%
       exposure        [0, 1]        0.08       15.69                    138             0.15%

========================= sev Outlier Statistics =========================
       Variable Defined Range  Actual Min  Actual Max  Out of Bounds (Count) Out of Bounds (%)
  equipment_age       [3, 25]        3.00      269.24                     46             0.56%
maintenance_int   [100, 2000]      100.01   21,980.46                      7             0.08%
      usage_int      [0, 100]        0.01      405.08                      8             0.10%
       exposure        [0, 1]        0.08       12.60                     11             0.13%
```

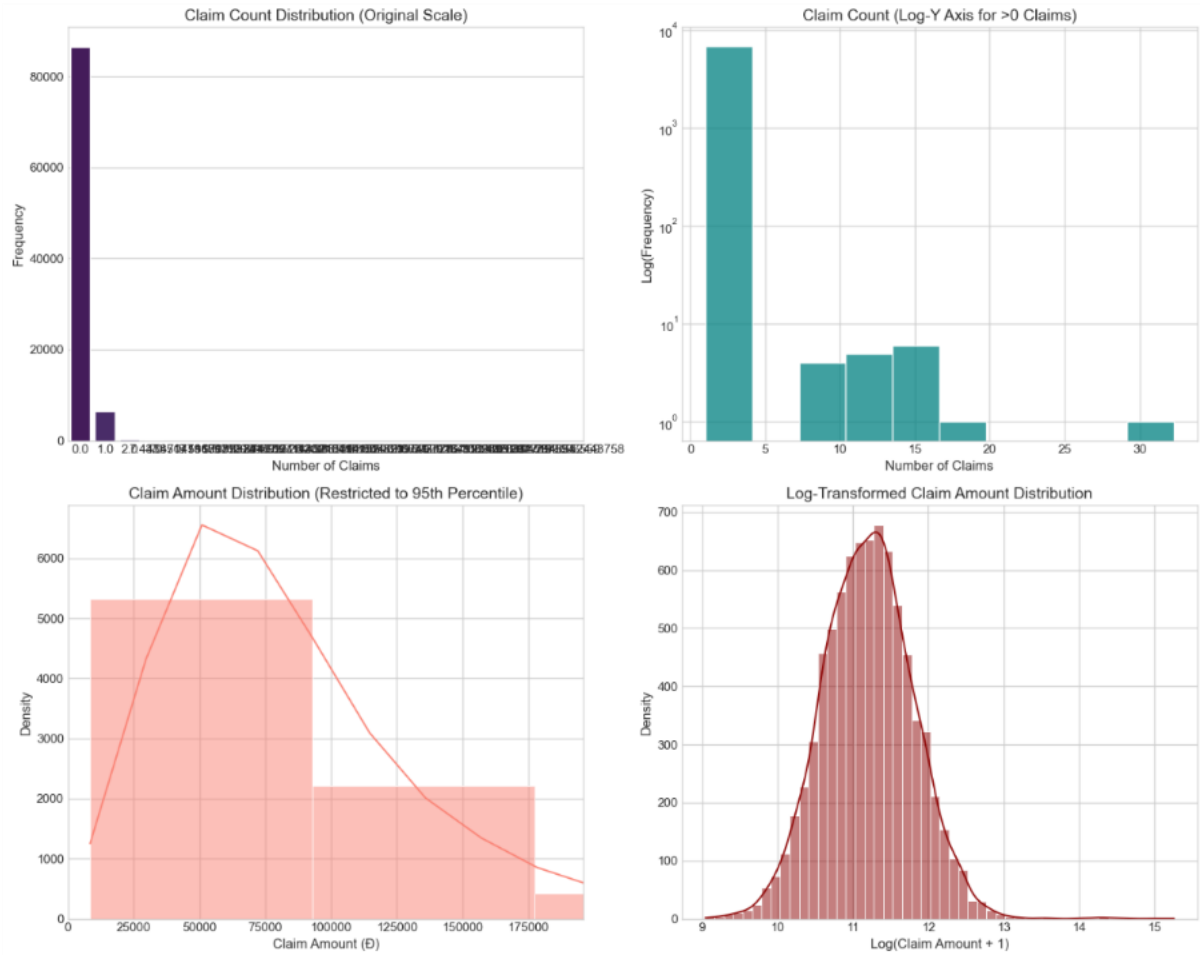
The image above shows the specific instances where each key metric exceeds the threshold. This helps us make decisions using the capping method.

Data range:

```
============================ Statistics for: freq ============================

[Numerical Variables]
                   count        mean      min      25%      50%      75%          max Defined Range
equipment_age    93421.0   10.208910    3.000    6.646   10.000   13.396    25.000000        3 - 25
maintenance_int  93421.0  953.261089  100.010  528.590  951.700 1379.160  2000.000000    100 - 2000
usage_int        93421.0   12.117688    0.000    6.050   12.020   18.000   100.000000       0 - 100
exposure         93421.0    0.500387    0.083    0.326    0.499    0.674     1.000000         0 - 1
claim_count      93421.0    0.080512    0.000    0.000    0.000    0.000    32.294355           N/A

[Categorical Variables (Top Categories)]
- equipment_type: Total Categories = 6
  * ReglAggregators: 26610 records
  * Quantum Bore: 22102 records
  * Ion Pulverizer: 17838 records
- solar_system: Total Categories = 3
  * Epsilon: 37345 records
  * Zeta: 37271 records
  * Helionis Cluster: 18805 records

============================ Statistics for: sev =============================

[Numerical Variables]
                  count          mean       min          25%        50%         75%           max Defined Range
claim_seq        8086.0      1.126232     1.000      1.00000      1.000       1.000  1.796942e+01           N/A
equipment_age    8086.0     11.649326     3.000      7.98175     11.650      15.034  2.500000e+01        3 - 25
maintenance_int  8086.0   1015.803444   100.010    606.68250   1046.850    1443.795  2.000000e+03    100 - 2000
usage_int        8086.0     14.222445     0.010      8.75000     15.100      20.010  1.000000e+02       0 - 100
exposure         8086.0      0.602683     0.083      0.46400      0.615       0.759  1.000000e+00         0 - 1
claim_amount     8086.0  89579.952484  8416.000  48510.50000  72967.000  108269.000  4.237030e+06           N/A

[Categorical Variables (Top Categories)]
- equipment_type: Total Categories = 6
  * ReglAggregators: 1845 records
  * Quantum Bore: 1695 records
  * Ion Pulverizer: 1586 records
- solar_system: Total Categories = 3
  * Epsilon: 3012 records
  * Zeta: 2754 records
  * Helionis Cluster: 2320 records
```

The image above shows the range of each variable. This serves as secondary confirmation of whether the capping method was successful.

Distribution:

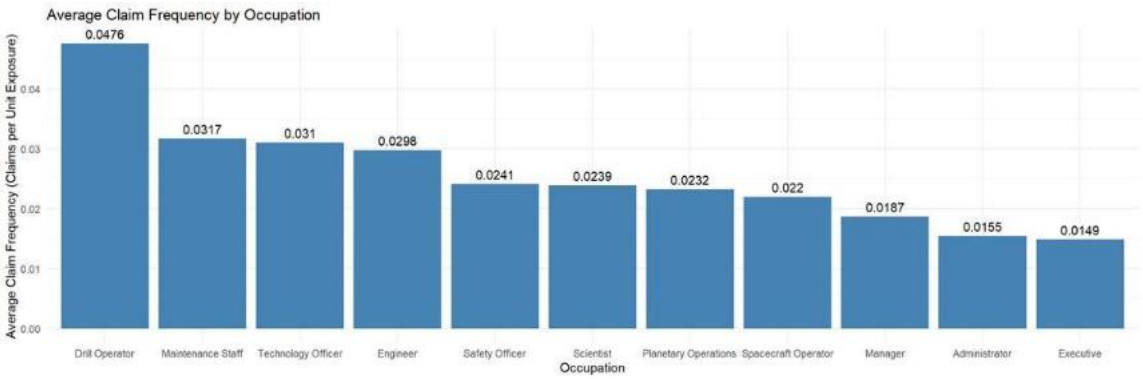


The figures above show the normal and log distributions of frequency and amount. This is in preparation for modelling

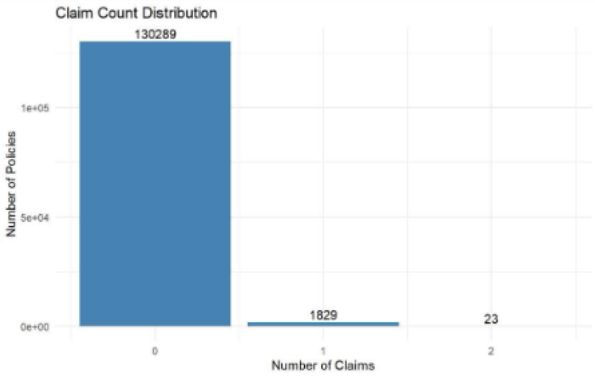
8.2.4 Workers' Compensation: Detailed Analysis of Covariates

The distribution of some variables and relationships between variables was shown by the following plots.

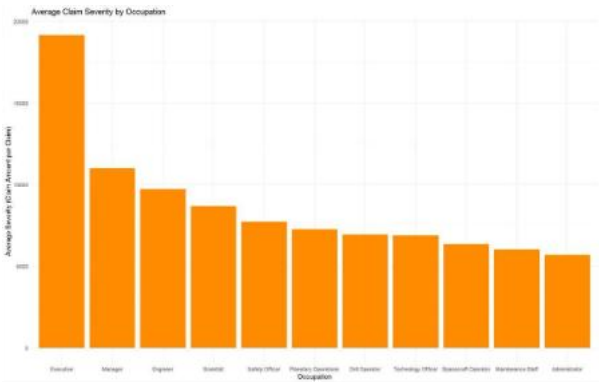
The plot below shows that there is a significant difference in claim frequency between occupations. Drill operators have the highest average claim frequency.



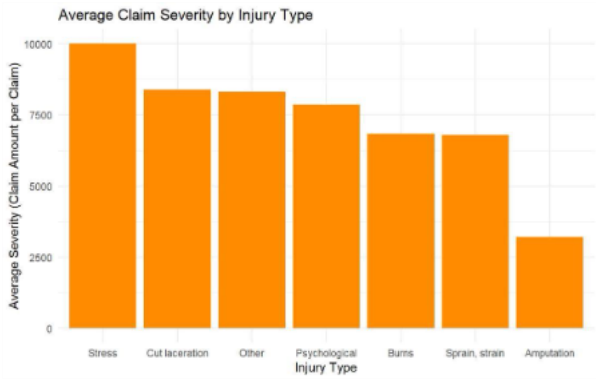
The plot below showed that most policies' claim count was 0, a small number of policies had 1 claim, and there were hardly any policies that had 2 claims.
The maximum claim count was 2.



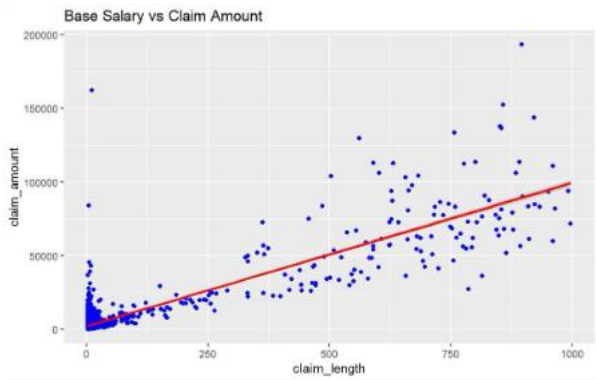
The plot below shows that there is a significant difference in claim severity by occupations. Executives have the highest average claim severity, caused by an outlier within only 10 data points.

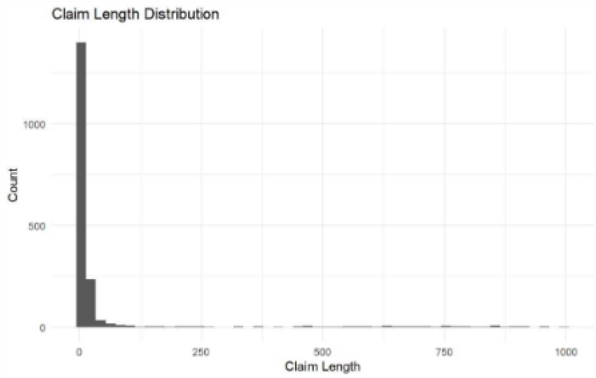


The plot below shows that there is a significant difference in claim severity by injury types. The highest average claim severity is caused by stress, which can lead to some mental illness.

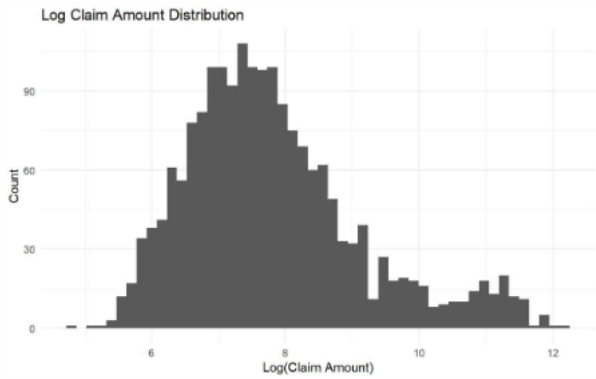


The plot below shows that the claim amount is increasing with claim length, and most claim lengths are short, but the tail is heavy, assuming a lot of outliers.





The plot below shows that the claim amount is right-tailed. Distributions like Gamma or lognormal can be used to fit it.



8.3 Product Design Appendices

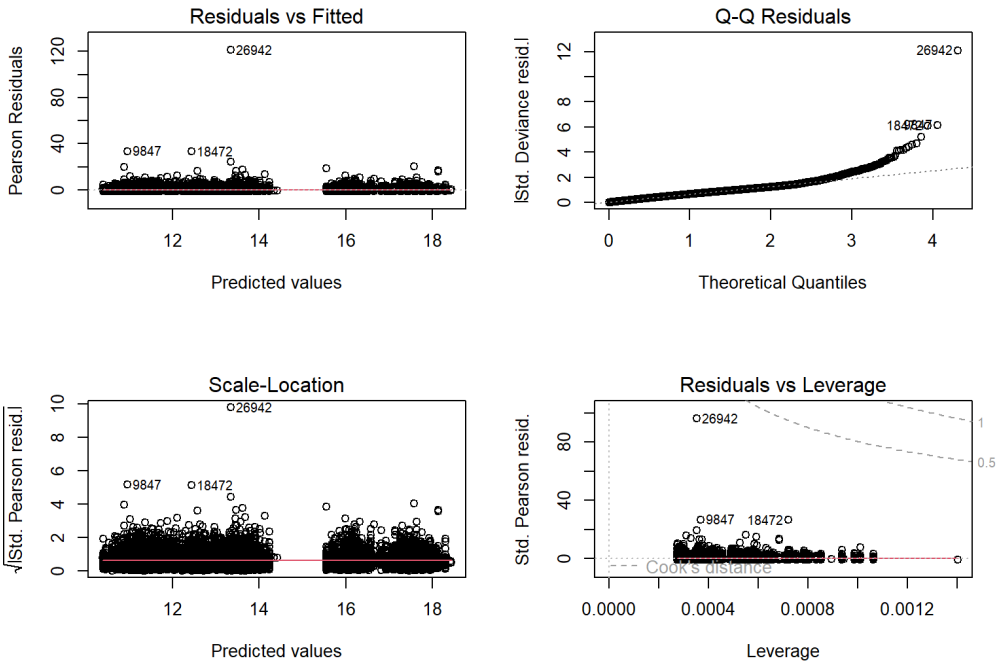
8.3.1 Business Interruption: Loss Measurement Benchmark

Item	**Description**
Primary reference	Exact budgeted output during the event downtime is available
Secondary reference	Agreed historical baseline is used when primary reference is unavailable
Threshold control	When no historical data is available, the loss shall not exceed the policy threshold

8.3.2 Cargo Loss: Relativity Calculations

Summary of Cargo Loss Severity Model and Fit

```
> summary(sev_glm)

Call:
glm(formula = claim_amount ~ cargo_type + route_risk + cargo_value +
    weight, family = Gamma(link = "log"), data = clms_cargo_sev)

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            1.180e+01  5.452e-02 216.456   <2e-16 ***
cargo_typegold         4.033e+00  9.583e-02  42.078   <2e-16 ***
cargo_typelithium      4.870e-01  2.941e-02  16.558   <2e-16 ***
cargo_typeplatinum     3.543e+00  5.576e-02  63.537   <2e-16 ***
cargo_typerare earths -3.346e-02  4.350e-02  -0.769    0.442
cargo_typesupplies    -1.784e+00  3.505e-02 -50.909   <2e-16 ***
cargo_typetitanium    -2.025e+00  2.906e-02 -69.700   <2e-16 ***
route_risk2            2.743e-01  2.954e-02   9.285   <2e-16 ***
route_risk3            4.882e-01  2.702e-02  18.065   <2e-16 ***
route_risk4            6.171e-01  2.910e-02  21.203   <2e-16 ***
route_risk5            8.293e-01  3.286e-02  25.238   <2e-16 ***
cargo_value            2.370e-09  1.593e-10  14.874   <2e-16 ***
weight                 7.277e-06  3.014e-07  24.146   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 1.586083)

    Null deviance: 155900  on 30605  degrees of freedom
Residual deviance:  20987  on 30593  degrees of freedom
AIC: 894551

Number of Fisher Scoring iterations: 5
```



Since the model uses a log link, the relationship between the claim amount and the variables Z and the coefficient estimates β is:

$$\ln(claim\ amount) = \sum \beta Z^T$$

$$claim\ amount = \prod e^{\beta Z^T}$$

This shows that the claim amounts would change by a multiplicative factor for changes in the variables used.

The risk relativities were then calculated in the table below.

Rating Factor	**Average/Total Coefficient** βZ^T	**Risk Relativity**
Transports gold or platinum	$\frac{4.033 + 3.543}{2} = 3.788$	$e^{3.788} = 44.17$ ≈ 44
Travels on route risk 4 or 5	$\frac{0.6171 + 0.8293}{2} = 0.7232$	$e^{0.7232} = 2.06$ ≈ 2
Transports more than 75 tonnes	$75000 \times 7.277 \times 10^{-6}$ $= 0.5458$	$e^{0.5458} = 1.726$ ≈ 1.75
Transports more than $90 million worth of cargo	$90000000 \times 2.37 \times 10^{-9}$ $= 0.2133$	$e^{0.2133} = 1.238$ ≈ 1.25

8.3.3 Equipment Failure: Rationale for Premium, Relativities, Deductibles/Exclusions

Rationale for Base Premium (Đ15,000)

GGIC set the base premium for equipment failure at Đ15,000. This figure is derived from the average cost per unit data across the primary mining zones. In the Helionis Cluster, the average cost per unit for various equipment types ranges from Đ10,703 to Đ16,779.

In the short term, the expected loss costs (pure premium) for most equipment hover around the Đ15,000 mark. In the long term, these costs increase due to machine aging and inflation. For example, the replacement cost for a Flux Rider in the Helionis Cluster reaches an average of Đ25,635 over a ten-year horizon. Therefore, Đ15,000 serves as a robust and stable baseline. When combined with risk relativities, this base premium provides sufficient coverage for high-risk assets, such as Ion Pulverizers in the Bayesia System, while remaining fair for lower-risk operations.

Rationale for Risk Relativities

Ion Pulverizer (1.30): Regression analysis shows a significant positive coefficient (+0.518) in the severity model for this equipment. These units contain high-energy ion beam components that are extremely expensive to repair. In the Bayesia System, the average loss for these units is 27% higher than the baseline.

Flux Rider (1.15): While the failure frequency is relatively low, long-term simulations show that Flux Riders have very high replacement costs, often exceeding Đ25,000. This relativity accounts for the physical wear and tear in high-mobility zones.

Equipment Age > 5 Years (1.20): Our logit regression model indicates that failure frequency spikes as equipment enters the later stages of its lifecycle (ages 15-20+). This multiplier addresses the mechanical fatigue and "bathtub curve" failure patterns.

No Claims Bonus (0.95): A 5% discount acts as an incentive for policyholders. This encourages proactive maintenance and reduces the overall administrative burden for GGIC.

System-Specific Deductibles and Exclusions

Different solar systems pose unique environmental challenges. GGIC uses specific deductibles to manage exposure to these catastrophic risks:

Solar System	Deductible(s)	Rationale
Bayesia System	Đ150,000 for radiation-related meltdowns	The binary stars create systemic radiation spikes. This deductible filters out minor circuit damage and focuses coverage on total system meltdowns.
Helionis Cluster	Đ100,000 for asteroid and debris damage	This region has the highest failure frequency (+0.39 coefficient). The deductible stabilizes premiums by sharing the cost of frequent minor impacts.

Oryn Delta	Đ250,000 for gravity-related failures	Gravitational fluctuations can cause irreversible structural warping and total loss. A high deductible ensures policyholders share the risk of these extreme events.

8.3.4 Workers' Compensation: Details About Allowable Expenses and Death Benefit

The medical expenses include hospital expenses, surgical costs, drug costs, rehabilitation therapy, physical therapy, and medical equipment costs.

The survivor benefit includes funeral expenses with a limit of Đ10000, which covers the cost of the funeral, cemetery, and other ceremonies. It also includes dependent benefits, such as 70% of the worker's salary, which will be paid weekly to the spouse or children of the worker until the spouse remarries or the children become adults.

The total premium for each solar system is calculated by the following formula.

$$P_i = \frac{\sum_{j \in i} E[P_j]}{\sum_{all\ j} E[P_j]} * P$$

Where $E[P_j]$ is the expected premium for each exposure, and P is the baseline premium. For Bayesia and Oryn Delta, assume their total exposures and risk levels remain the same with Epsilon and Zeta as a conservative estimation.

The cost of reinsurance, although considered a way to cover extreme loss, is not included in the modelling part.

8.4 Pricing and Capital Modelling Appendices

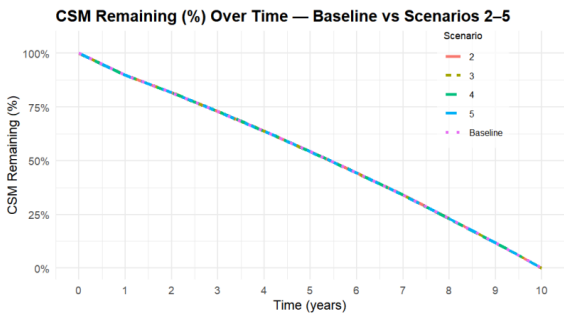
8.4.1 Business Interruption

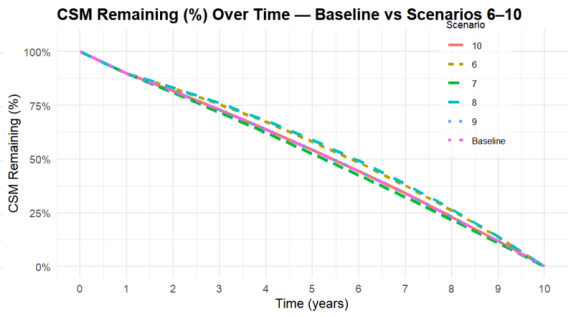
Scenario	Expense Ratio	CoC Rate	Investment Return Rate	1-Year Risk-free Rate	Inflation Rate	Discontinuance rates
Baseline	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	0
2	45%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	0
3	60%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	0
4	30%	1-Year Risk-free Rate+8%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	0
5	30%	1-Year Risk-free Rate+10%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	0
6	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	5.45%	2.465%	0
7	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	0.15%	2.465%	0
8	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	1.79%	7.08%	0
9	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate-1%	1.79%	2.465%	0
10	30%	1-Year Risk-free Rate+6%	1-Year Risk-free Rate+0.5%	1.79%	2.465%	10%

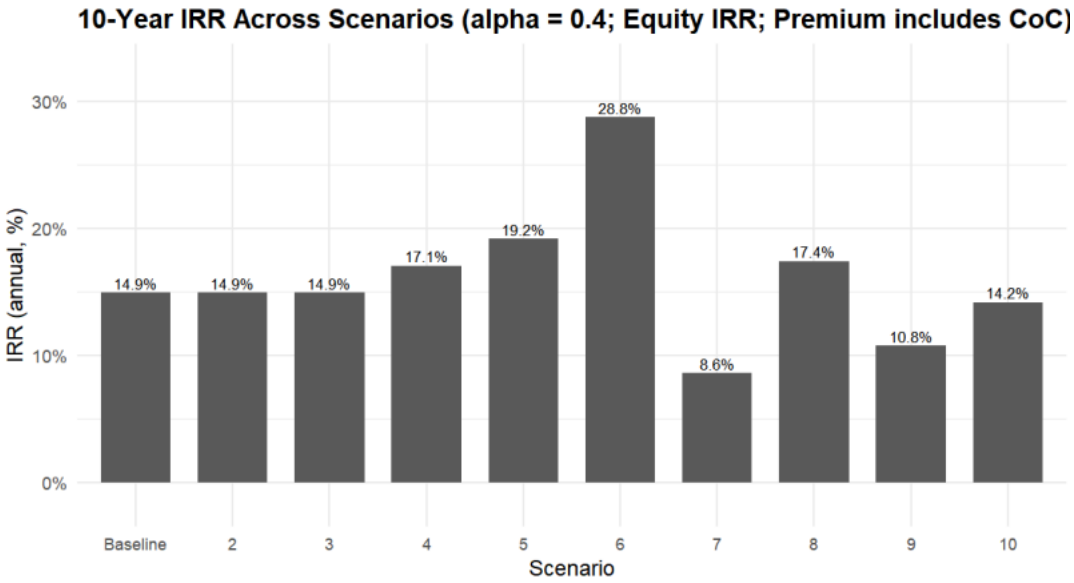
Ten different scenarios with different set-up of expense ratio, CoC rate, investment return, interest rate, inflation rate and discontinuance rates are simulated and tested for their CSM (Contractual Service Margin) and IRR performance. 30% of expense ratio, which is considerably higher than that in general assumptions and reality, is intentionally used for variety of extreme conditions, but shows no influence on the CSM and IRR.

When applying discontinuance rates, we assume that each policy has the same probability of 10% to lapse half-a-year on average after the beginning of the contract.

When lapse happens, we assume the company have enough cash or reserve to return the unearned premium, neglecting the extra management fee. In such way, the investment won't be affected.







The 10-year equity IRR is mostly in the mid-teens around 14–19% across most scenarios, and the overall profitability level is fairly stable.

Scenario 6 stands out with the highest IRR about 29%. This is consistent with that scenario assuming a higher annual investment return, which increases investment income over the 10-year horizon and lifts the equity IRR noticeably above the baseline.

8.4.2 Cargo Loss

8.4.2.1 Base Premium Calculation

Summary of Claim Frequency Model Using Negative Binomial GLM with Offset Term

```
Call:
glm.nb(formula = claim_count ~ route_risk + pilot_experience +
    debris_density + offset(log(exposure)), data = clms_cargo_freq,
    init.theta = 0.1499075107, link = log)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)      -0.282781   0.041488  -6.816 9.36e-12 ***
route_risk2       0.103214   0.034333   3.006 0.002644 **
route_risk3       0.122372   0.031664   3.865 0.000111 ***
route_risk4       0.161376   0.035001   4.611 4.02e-06 ***
route_risk5       0.168175   0.040602   4.142 3.44e-05 ***
pilot_experience -0.003875   0.001758  -2.204 0.027514 *
debris_density    0.150372   0.083519   1.800 0.071790 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(0.1499) family taken to be 1)

    Null deviance: 62382  on 124751  degrees of freedom
Residual deviance: 62351  on 124745  degrees of freedom
AIC: 178236

Number of Fisher Scoring iterations: 1


              Theta:  0.14991
          Std. Err.:  0.00177

 2 x log-likelihood:  -178220.17100
```

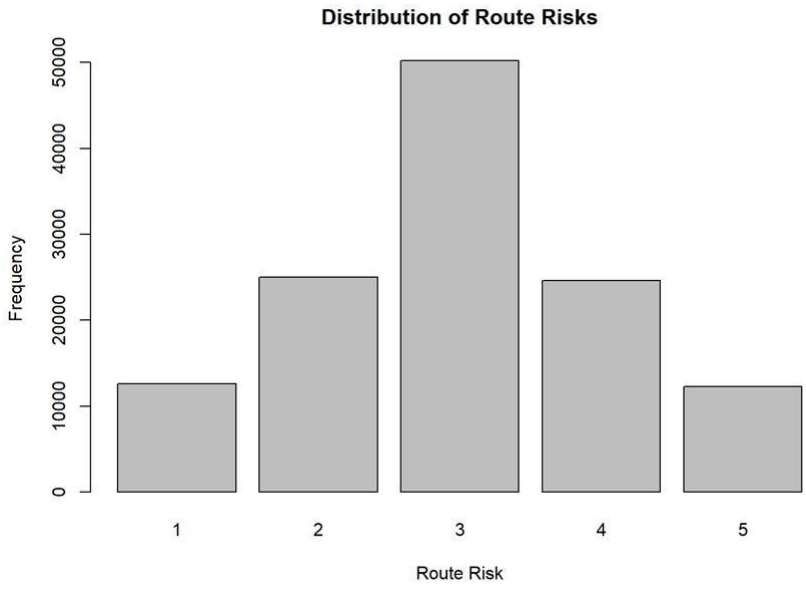
This negative binomial model uses a log link and, hence the claim count can be calculated with respect to the model's variables Z and coefficient estimates β.

$$claim\ count = \prod e^{\beta Z^T}$$

The base premium was set by using the intercepts of both the frequency and severity (from Appendix 8.3.2) models.

$$base = e^{-0.282781+11.8} = 100430.2765$$

Furthermore, it can be observed that there are an extremely large number of policyholders in a route risk of category 3.



As a result, the category 3 route risk estimate from the severity model was incorporated into the base premium.

$$base = 100430.2765e^{0.4882} = 163639.1537$$

Finally, a loading of 20% was incorporated which included operational costs, regulatory costs (e.g. taxes), reinsurer loading and profit loading.

$$base = 163639.1537 \times 1.2 = \$196421.30$$

8.4.2.2 Stress Test Results

All route risks are category 5

Short-term	**Claims**	**Premiums**	**Net Revenue**
Expected Value	399 billion	491 billion	52 billion
Variance	3.188×10^{21}	4.776×10^{21}	8.796×10^{19}
VaR 99%	565 billion	370 billion	33 billion
TVaR 99%	607 billion	356 billion	31 billion
Long-term			
Expected Value	4.366 trillion	5.510 trillion	685 billion
Variance	1.587×10^{24}	2.548×10^{24}	4.188×10^{22}
VaR 99%	8.377 trillion	2.990 trillion	361 billion
TVaR 99%	9.533 trillion	2.794 trillion	336 billion

All debris densities are maximised i.e. equal to 1

Short-term	**Claims**	**Premiums**	**Net Revenue**
Expected Value	428 billion	493 billion	23 billion
Variance	3.621×10^{21}	4.748×10^{21}	5.291×10^{19}
VaR 99%	603 billion	369 billion	7.694 billion
TVaR 99%	649 billion	357 billion	5.561 billion
Long-term			
Expected Value	4.724 trillion	5.557 trillion	384 billion
Variance	1.835×10^{24}	2.577×10^{24}	1.427×10^{22}
VaR 99%	9.021 trillion	2.983 trillion	192 billion
TVaR 99%	10.115 trillion	2.768 trillion	178 billion

All route risks are category 5 and debris densities are maximised (i.e. completely catastrophic event)

Short-term	**Claims**	**Premiums**	**Net Revenue**
Expected Value	448 billion	493 billion	3.590 billion
Variance	3.949×10^{21}	4.733×10^{21}	4.364×10^{19}
VaR 99%	638 billion	367 billion	(11) billion
TVaR 99%	687 billion	356 billion	(14) billion
Long-term			
Expected Value	4.913 trillion	5.542 trillion	167 billion
Variance	1.993×10^{24}	2.558×10^{24}	3.596×10^{21}
VaR 99%	9.438 trillion	3.048 trillion	72 billion
TVaR 99%	10.618 trillion	2.848 trillion	65 billion

8.4.3 Equipment Failure

8.4.3.1 Frequency and Severity Model Details & Revenue Calculations

Frequency Modelling:

We used a Poisson distribution for the number of claims (N):

N ~ Poisson(λ_adj)

The adjusted frequency (λ_adj) is calculated as:

λ_adj = Freq_Prob × Count × Risk_Factor × Mean_Intensity × Shared_Shock

Where Freq_Prob is derived from a logistic link (Logit):

Logit = β_0 + β_age × Age + β_usage × Usage_Std + β_maint × Maint_Std

Freq_Prob = 1 / (1 + e^(-Logit))

Code output:

```
============================= Frequency Model Step 1: Logit ==============================
                           Logit Regression Results
==============================================================================
Dep. Variable:              has_claim   No. Observations:                93421
Model:                          Logit   Df Residuals:                    93406
Method:                           MLE   Df Model:                           14
Date:                Fri, 13 Mar 2026   Pseudo R-squ.:                 0.05902
Time:                        17:29:08   Log-Likelihood:                -23174.
converged:                       True   LL-Null:                       -24627.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================================
                                         coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
Intercept                             -1.3384      0.057    -23.327      0.000      -1.451      -1.226
C(equipment_type)[T.Flux Rider]       -0.1063      0.060     -1.760      0.078      -0.225       0.012
C(equipment_type)[T.Graviton Extractor]  -0.3789      0.059     -6.462      0.000      -0.494      -0.264
C(equipment_type)[T.Ion Pulverizer]   -0.4538      0.057     -7.993      0.000      -0.565      -0.343
C(equipment_type)[T.Quantum Bore]     -0.5850      0.056    -10.437      0.000      -0.695      -0.475
C(equipment_type)[T.ReglAggregators]  -0.7077      0.055    -12.755      0.000      -0.816      -0.599
C(age_bin)[T.15-19]                    0.2964      0.036      8.157      0.000       0.225       0.368
C(age_bin)[T.20+]                      0.6176      0.067      9.207      0.000       0.486       0.749
C(age_bin)[T.5-9]                     -0.2505      0.031     -7.998      0.000      -0.312      -0.189
C(age_bin)[T.<5]                      -0.5271      0.044    -11.969      0.000      -0.613      -0.441
C(solar_system)[T.Helionis Cluster]    0.3972      0.032     12.308      0.000       0.334       0.460
C(solar_system)[T.Zeta]               -0.0866      0.030     -2.919      0.004      -0.145      -0.028
usage_int_std                          0.2271      0.011     20.599      0.000       0.206       0.249
maintenance_int_std                    0.1210      0.013      9.492      0.000       0.096       0.146
log_exposure                           1.0222      0.030     34.483      0.000       0.964       1.080
==================================================================================================
Empirical Mean Intensity: 1.0894
```

Severity Modelling:

We applied Extreme Value Theory (EVT) to handle heavy-tailed risks:

* Body (Below Threshold): For losses ≤ 197,906 Đ, we used a Gamma Distribution:

f(x) ~ Gamma(α = 1/φ, β = μ × φ)

* Tail (Above Threshold): For losses > 197,906 Đ, we used a Generalized Pareto Distribution (GPD):

G(x; ξ, σ) = 1 - (1 + ξ × (x - u) / σ)^(-1/ξ)

* Profit Margin: 5.0%

* System Multipliers (Risk Load):

* Helionis: 1.25 | Bayesia: 1.35 | Oryn: 1.45(Based on the galactic environment)

Code output:

```
============================= Severity Model: Gamma Body =============================
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:           claim_amount   No. Observations:                 7695
Model:                            GLM   Df Residuals:                     7681
Model Family:                   Gamma   Df Model:                           13
Link Function:                    Log   Scale:                          0.20364
Method:                          IRLS   Log-Likelihood:                -90587.
Date:                Fri, 13 Mar 2026   Deviance:                       1555.0
Time:                        17:29:08   Pearson chi2:                 1.56e+03
No. Iterations:                     13   Pseudo R-squ. (CS):             0.2699
Covariance Type:            nonrobust
==============================================================================================
                                          coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                               10.7621      0.022    492.260      0.000      10.719      10.805
C(equipment_type)[T.Flux Rider]          0.1980      0.023      8.705      0.000       0.153       0.243
C(equipment_type)[T.Graviton Extractor]  0.3997      0.022     17.942      0.000       0.356       0.443
C(equipment_type)[T.Ion Pulverizer]      0.5184      0.021     24.124      0.000       0.476       0.561
C(equipment_type)[T.Quantum Bore]        0.7897      0.022     36.446      0.000       0.747       0.832
C(equipment_type)[T.ReglAggregators]     0.4773      0.021     22.659      0.000       0.436       0.519
C(age_bin)[T.15-19]                      0.0577      0.014      4.038      0.000       0.030       0.086
C(age_bin)[T.20+]                        0.1225      0.025      4.845      0.000       0.073       0.172
C(age_bin)[T.5-9]                       -0.0472      0.013     -3.679      0.000      -0.072      -0.022
C(age_bin)[T.<5]                        -0.0988      0.019     -5.329      0.000      -0.135      -0.062
C(solar_system)[T.Helionis Cluster]     -0.0810      0.013     -6.379      0.000      -0.106      -0.056
C(solar_system)[T.Zeta]                  0.0936      0.012      7.632      0.000       0.070       0.118
usage_int_std                            0.1274      0.005     24.300      0.000       0.117       0.138
log_exposure                             0.0258      0.012      2.065      0.039       0.001       0.050
==============================================================================================
GPD Parameter Fit Results: Shape(xi)=0.5213, Scale(sigma)=46861.59
```

Pricing and Revenue Formulas

The following formulas were used to determine the financial metrics per equipment unit and for the total portfolio:

Expected Cost (Exp Cost):

Exp Cost = E[∑(i=1 to N) X_i]

(Mean of all simulated total loss paths)

Expected Revenue/Premium (Exp Ret):

Exp Ret = (Exp Cost × System Multiplier) / (1 - Profit Margin Rate)

Net Revenue (Exp Net):

Exp Net = Exp Ret × Profit Margin Rate

8.4.3.2 Stress Testing Results

For systemic risk analysis, we incorporated Shared Cosmic Shocks using Gamma distributions (α=6.67, β=0.15).

* Helionis: 90% exposure to debris shocks.

* Bayesia: 90% exposure to radiation shocks.

* Worst-Case Scenario: A simulated systemic catastrophe (Frequency +100% and shared dependency active), used to set the upper bound for the Tail-Risk Pool.

Code output:

Short Term (1 Year) Report:

```
============================ Short Term (1 Year) Report ============================

[System-Level Summary - Aggregate Data]
System                | Exp Cost | Var Cost | VaR Cost | Exp Ret | Var Ret | VaR Ret | Exp Net | Var Net | VaR Net
Helionis Cluster      | 40,567,876 |  7.4e+12 | 47,331,822 | 53,378,785 |  5.6e-17 | 53,378,785 | 2,668,939 |  0.0e+00 | 2,668,939
Bayesia System        | 14,826,272 |  2.5e+12 | 19,149,518 | 21,068,913 |  1.4e-17 | 21,068,913 | 1,053,446 |  5.4e-20 | 1,053,446
Oryn Delta            |  6,989,922 |  1.3e+12 |  9,896,074 | 10,668,828 |  3.5e-18 | 10,668,828 |   533,441 |  1.4e-20 |   533,441

[Equipment-Level Detail - Average Per Unit Data]

--- Helionis Cluster ---
Quantum Bore          | Avg Cost:     14,817 | Avg Ret:     19,496 | Avg Net:        975
Graviton Extractor    | Avg Cost:     14,204 | Avg Ret:     18,689 | Avg Net:        934
FexStram Carrier      | Avg Cost:     13,040 | Avg Ret:     17,158 | Avg Net:        858
ReglAggregators       | Avg Cost:     10,703 | Avg Ret:     14,083 | Avg Net:        704
Flux Rider            | Avg Cost:     16,779 | Avg Ret:     22,077 | Avg Net:      1,104
Ion Pulverizer        | Avg Cost:     15,437 | Avg Ret:     20,312 | Avg Net:      1,016

--- Bayesia System ---
Quantum Bore          | Avg Cost:      9,702 | Avg Ret:     13,787 | Avg Net:        689
Graviton Extractor    | Avg Cost:      9,136 | Avg Ret:     12,983 | Avg Net:        649
FexStram Carrier      | Avg Cost:      8,521 | Avg Ret:     12,109 | Avg Net:        605
ReglAggregators       | Avg Cost:      8,636 | Avg Ret:     12,273 | Avg Net:        614
Flux Rider            | Avg Cost:     12,156 | Avg Ret:     17,275 | Avg Net:        864
Ion Pulverizer        | Avg Cost:     19,142 | Avg Ret:     27,201 | Avg Net:      1,360

--- Oryn Delta ---
Quantum Bore          | Avg Cost:      9,518 | Avg Ret:     14,528 | Avg Net:        726
Graviton Extractor    | Avg Cost:      9,102 | Avg Ret:     13,893 | Avg Net:        695
FexStram Carrier      | Avg Cost:      5,974 | Avg Ret:      9,118 | Avg Net:        456
ReglAggregators       | Avg Cost:      8,008 | Avg Ret:     12,223 | Avg Net:        611
Flux Rider            | Avg Cost:      7,340 | Avg Ret:     11,202 | Avg Net:        560
Ion Pulverizer        | Avg Cost:     14,579 | Avg Ret:     22,252 | Avg Net:      1,113
```

Long Term (10 Years) Report:

```
============================ Long Term (10 Years) Report ============================

[System-Level Summary - Aggregate Data]
System                | Exp Cost |  Var Cost |  VaR Cost |  Exp Ret |  Var Ret |  VaR Ret |  Exp Net |  Var Net |  VaR Net
Helionis Cluster      | 70,714,592 |  1.6e+13 | 80,358,418 | 93,045,516 |  2.2e-16 | 93,045,516 |  4,652,276 |  0.0e+00 |  4,652,276
Bayesia System        | 26,759,946 |  1.0e+13 | 32,850,937 | 38,027,292 |  2.2e-16 | 38,027,292 |  1,901,365 |  2.2e-19 |  1,901,365
Oryn Delta            | 12,080,222 |  2.7e+12 | 16,255,822 | 18,438,233 |  1.4e-17 | 18,438,233 |    921,912 |  2.2e-19 |    921,912

[Equipment-Level Detail - Average Per Unit Data]

--- Helionis Cluster ---
Quantum Bore          | Avg Cost:     18,954 | Avg Ret:     24,939 | Avg Net:      1,247
Graviton Extractor    | Avg Cost:     17,575 | Avg Ret:     23,125 | Avg Net:      1,156
FexStram Carrier      | Avg Cost:     14,282 | Avg Ret:     18,792 | Avg Net:        940
ReglAggregators       | Avg Cost:     15,101 | Avg Ret:     19,869 | Avg Net:        993
Flux Rider            | Avg Cost:     25,635 | Avg Ret:     33,730 | Avg Net:      1,686
Ion Pulverizer        | Avg Cost:     17,145 | Avg Ret:     22,560 | Avg Net:      1,128

--- Bayesia System ---
Quantum Bore          | Avg Cost:     12,954 | Avg Ret:     18,409 | Avg Net:        920
Graviton Extractor    | Avg Cost:     11,771 | Avg Ret:     16,727 | Avg Net:        836
FexStram Carrier      | Avg Cost:      9,813 | Avg Ret:     13,944 | Avg Net:        697
ReglAggregators       | Avg Cost:     12,399 | Avg Ret:     17,620 | Avg Net:        881
Flux Rider            | Avg Cost:     19,357 | Avg Ret:     27,507 | Avg Net:      1,375
Ion Pulverizer        | Avg Cost:     20,868 | Avg Ret:     29,654 | Avg Net:      1,483

--- Oryn Delta ---
Quantum Bore          | Avg Cost:     12,641 | Avg Ret:     19,294 | Avg Net:        965
Graviton Extractor    | Avg Cost:     11,901 | Avg Ret:     18,165 | Avg Net:        908
FexStram Carrier      | Avg Cost:      6,814 | Avg Ret:     10,401 | Avg Net:        520
ReglAggregators       | Avg Cost:     11,784 | Avg Ret:     17,986 | Avg Net:        899
Flux Rider            | Avg Cost:     12,557 | Avg Ret:     19,165 | Avg Net:        958
Ion Pulverizer        | Avg Cost:     16,364 | Avg Ret:     24,976 | Avg Net:      1,249

=================== Tail Risk Analysis (TVaR 99% System Total) ===================
- Helionis Cluster: Short Term 51,642,229 | Long Term 89,503,429
- Bayesia System: Short Term 21,500,477 | Long Term 40,145,179
- Oryn Delta: Short Term 12,064,533 | Long Term 19,604,235
```

Stress Test Results

```
##############################################################
# 4. Stress and Scenario Testing (n=2000)
##############################################################
-------------------------------------------------------------------------------------
Scenario                        |    Helionis VaR |     Bayesia VaR |        Oryn VaR
-------------------------------------------------------------------------------------
Baseline (Ref)                  |      46,841,225 |      18,808,033 |       9,716,719
Freq Stress (+50%)              |      71,841,497 |      27,126,315 |      14,136,756
Inflation Stress (+5%)          |      48,565,259 |      19,737,636 |      10,314,297
Dependency (Risk-Based)         |      85,118,848 |      31,104,545 |      13,177,898
Best-Case (Smooth Ops)          |      38,526,826 |      15,791,969 |       8,092,599
Moderate-Case (Batch Defect)    |      62,094,048 |      24,195,641 |      12,296,274
Worst-Case (Solar Flares)       |     168,109,180 |      59,129,826 |      26,113,038
```

8.4.4 Workers' Compensation

8.4.4.1 Frequency and Severity Model Details

The summary of the Poisson GLM frequency model is:

```
Call:
glm(formula = claim_count ~ occupation + accident_history_flag +
    psych_stress_index + safety_training_index + offset(log(exposure)),
    family = poisson(), data = freqc)

Coefficients:
                              Estimate Std. Error z value Pr(>|z|)
(Intercept)                   -4.02266    0.13309 -30.225  < 2e-16 ***
occupationDrill Operator       1.12521    0.10582  10.633  < 2e-16 ***
occupationEngineer             0.65812    0.11415   5.766 8.14e-09 ***
occupationExecutive           -0.04542    0.33109  -0.137 0.890882
occupationMaintenance Staff    0.71684    0.14623   4.902 9.48e-07 ***
occupationManager              0.19079    0.12244   1.558 0.119194
occupationPlanetary Operations 0.40611    0.12709   3.195 0.001396 **
occupationSafety Officer       0.44810    0.12596   3.557 0.000374 ***
occupationScientist            0.43545    0.14979   2.907 0.003648 **
occupationSpacecraft Operator  0.35518    0.15331   2.317 0.020522 *
occupationTechnology Officer   0.69572    0.13936   4.992 5.96e-07 ***
accident_history_flag1         0.25744    0.06913   3.724 0.000196 ***
psych_stress_index2            0.05437    0.06279   0.866 0.386517
psych_stress_index3            0.21446    0.06546   3.276 0.001053 **
psych_stress_index4            0.33034    0.07348   4.495 6.95e-06 ***
psych_stress_index5            0.23044    0.10303   2.237 0.025309 *
safety_training_index2        -0.11305    0.11154  -1.014 0.310810
safety_training_index3        -0.15727    0.10498  -1.498 0.134110
safety_training_index4        -0.30267    0.10322  -2.932 0.003365 **
safety_training_index5        -0.36799    0.09514  -3.868 0.000110 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 15575  on 132140  degrees of freedom
Residual deviance: 15255  on 132121  degrees of freedom
AIC: 19013

Number of Fisher Scoring iterations: 7
```

The summary of the Gamma GLM severity model is:

```
Call:
glm(formula = claim_amount ~ solar_system + station_id + employment_type +
    psych_stress_index + supervision_level + gravity_level +
    safety_training_index + base_salary, family = Gamma(link = "log"),
    data = sevc)

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  1.004e+01  7.476e-01  13.436  < 2e-16 ***
solar_systemHelionis Cluster 7.635e-01  2.395e-01   3.188 0.001457 **
solar_systemZeta             1.002e+00  3.920e-01   2.557 0.010644 *
station_idA2                -2.532e-01  4.000e-01  -0.633 0.526777
station_idA3                -6.445e-01  3.710e-01  -1.737 0.082494 .
station_idA4                -8.001e-01  3.950e-01  -2.025 0.042965 *
station_idA5                -3.578e-01  3.901e-01  -0.917 0.359234
station_idA6                -2.220e-01  4.021e-01  -0.552 0.580913
station_idA7                -7.169e-01  4.175e-01  -1.717 0.086151 .
station_idA8                -5.968e-01  3.852e-01  -1.549 0.121457
station_idA9                -7.274e-01  3.815e-01  -1.906 0.056758 .
station_idB1                -6.271e-01  3.988e-01  -1.573 0.115981
station_idB2                -8.029e-01  3.832e-01  -2.095 0.036292 *
station_idB3                -8.909e-01  3.882e-01  -2.295 0.021836 *
station_idB4                -3.236e-01  3.733e-01  -0.867 0.386106
station_idB5                -5.443e-01  3.873e-01  -1.405 0.160068
station_idB6                -4.725e-01  3.941e-01  -1.199 0.230697
station_idB7                -3.593e-02  3.661e-01  -0.098 0.921822
station_idB8                -6.435e-01  3.849e-01  -1.672 0.094745 .
station_idB9                -3.307e-01  3.922e-01  -0.843 0.399249
station_idG1                -1.567e-01  4.206e-01  -0.373 0.709522
station_idG2                -8.027e-01  3.682e-01  -2.180 0.029371 *
station_idG3                -5.261e-01  3.943e-01  -1.334 0.182344
station_idG4                -2.487e-01  3.701e-01  -0.672 0.501730
station_idG5                -8.661e-01  3.939e-01  -2.199 0.028022 *
station_idG6                -5.499e-01  3.741e-01  -1.470 0.141676
station_idG7                -7.290e-01  3.898e-01  -1.870 0.061610 .
station_idG8                -6.541e-01  3.769e-01  -1.735 0.082852 .
station_idG9                -9.681e-01  3.624e-01  -2.671 0.007632 **
employment_typeFull-time     3.255e-01  1.116e-01   2.916 0.003586 **
psych_stress_index2          4.560e-01  1.434e-01   3.180 0.001498 **

psych_stress_index3          4.509e-01  1.498e-01   3.010 0.002649 **
psych_stress_index4          5.852e-01  1.681e-01   3.482 0.000510 ***
psych_stress_index5          8.034e-01  2.376e-01   3.381 0.000738 ***
supervision_level           -6.214e-01  3.236e-01  -1.920 0.054977 .
gravity_level               -1.811e+00  7.408e-01  -2.444 0.014611 *
safety_training_index2      -4.058e-01  2.580e-01  -1.573 0.115866
safety_training_index3      -4.867e-01  2.423e-01  -2.009 0.044698 *
safety_training_index4      -5.213e-01  2.384e-01  -2.186 0.028919 *
safety_training_index5      -6.008e-01  2.199e-01  -2.733 0.006345 **
base_salary                  2.054e-05  3.570e-06   5.754 1.02e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 5.114408)

    Null deviance: 4323.7  on 1864  degrees of freedom
Residual deviance: 3822.1  on 1824  degrees of freedom
AIC: 36435

Number of Fisher Scoring iterations: 12
```

The costs are simulated based on an exposure of 132,141. The lambda (mean frequency) is 1,875, and the mean severity is 8,063.

The returns are calculated by the formula below, which considers the expense, risk level, and profit margin. The "Costs" are defined as short-term loss.

$$Returns = \frac{E[Costs] + Risk\ load + Expense\ load}{1 - Profit\ margin}$$

The risk load is calculated by the following formula:

$$Risk\ load = (VaR95(Costs) - E[Costs]) * Risk\ factor$$

The risk factor is set to be 0.1, the expense load is set to be 0.2, and the profit margin is set to be 0.05.

8.4.4.2 Stress and Scenario Testing Results

Stress Testing

Short-term operating results of the workers' compensation insurance are shown below. The first case is an extreme claim frequency (1.5 times baseline) with a baseline severity:

	Costs	Returns	Net Revenue
Mean	31.53 million	26.89 million	(4.64 million)
Variances	5.18 trillion	0	5.18 trillion
VaR99	37.44 million	-	(10.55 million)
TVaR99	38.47 million	-	(11.58 million)

The second case is an extreme claim severity (1.8 times baseline) with a baseline frequency:

	Costs	Returns	Net Revenue
Mean	37.83 million	26.89 million	(10.94 million)
Variances	11.20 trillion	0	11.20 trillion
VaR99	46.65 million	-	(19.76 million)
TVaR99	48.25 million	-	(21.35 million)

The last case is a catastrophe with both extreme claim frequency and severity:

	Costs	Returns	Net Revenue
Mean	56.75 million	26.89 million	(29.85 million)
Variances	16.79 trillion	0	16.79 trillion
VaR99	67.39 million	-	(40.50 million)
TVaR99	69.25 million	-	(42.35 million)

Although both extreme scenarios above give a negative profit, they are unlikely to happen.

Scenario Testing

Under the best scenario, with both low claim frequency (0.9 times baseline) and severity (0.95 times baseline):

	Costs	Returns	Net Revenue
Mean	17.98 million	26.89 million	8.92 million
Variances	2.81 trillion	0	2.81 trillion
VaR99	22.45 million	-	4.44 million
TVaR99	23.27 million	-	3.62 million

Under the worst scenario, with both high claim frequency (1.2 times baseline) and severity (1.3 times baseline):

	Costs	Returns	Net Revenue
Mean	32.79 million	26.89 million	(5.89 million)
Variances	7.04 trillion	0	7.04 trillion
VaR99	39.68 million	-	(12.78 million)
TVaR99	40.86 million	-	(13.96 million)

Workers' compensation is no longer profitable under and beyond the worst-case scenario. A reinsurance policy should be applied to ensure the profitability of the product.

8.5 Assumptions Appendices

Business Interruption

- Exposure follows Normal distribution with mean of 0.5.
- The total exposure of all policy in the whole year should be the same as in the data, representing unchanged insured volume.
- No data in claim amount is removed, since no data exceed Q3+3IQR.
- Require capital or solvency capital is equal to VaR99%-Expected loss.
- Expenses are modelled as a fixed expense ratio of 10% of premium and assumed to be incurred at mid-year in moderate scenarios.
- Inflation (2.39%) and investment income are not applied in the 1-year PV calculation.
- In long-term simulation, we assume that 40% of net premiums (after expenses) is held as investable float on average.

Cargo Loss

Metric	Description	Rationale
Cargo and Vessel Durability	All cargo and container vessels can endure all climates and temperatures of all planets in each solar system.	Due to the extreme variations of climates and temperatures, it is assumed that space vessels and their cargo must be able to endure these conditions to be transported.
Route Database	All cargo travel on routes that have been established and recorded into an intergalactic database.	These routes have been established as they have been shown to maximise efficiency and safety, whilst reducing costs.
Claim Timing	Claims are made immediately upon completion of the transportation process. If the cargo does not reach its destination, the information about the registered route from the intergalactic route database and the cargo being transported will be used.	The Interplanetary Claims Grid only exists on the surface of planets. As a result, damage can only be inspected and evaluated at the end of the transit. This will then apply information about the registered route, which includes the solar systems it traverses through, and the cargo to determine the claim amount.
Claim Frequency	The distribution of the claim frequency for each independent policyholder is a Negative Binomial distribution based on their risk profile.	The variance is larger than the mean, and certain characteristics were shown to inflate higher claim frequencies.
Claim Severity	The distribution of the claim severity for each independent policyholder is a Gamma distribution based on their risk profile.	The historical data of severities was bimodal and there were correlations between certain risk characteristics and claim amounts.

Equipment Failure

- Financial & Timing: We use a "mid-point" approach for payments, assuming claims occur steadily throughout the year. Costs are discounted using 2074 spot rates (4.74%–5.10%) to reflect the time value of money.
- Equipment Degradation: We assume hardware failure rates rise steadily over the 10-year term. This captures the natural "wear and tear" of operating sensitive machinery in high-stress space environments.
- Cosmic Risk Correlation: While individual failures are often random, we include "shared shocks"—like solar flares or debris clouds—that can damage multiple units at once. This ensures our capital reserves can handle system-wide catastrophes.
- Streamlined Settlements: We assume all claims are filed and paid instantly via the Interplanetary Claims Grid. This removes bureaucratic lag and focuses our model on actual loss volatility.
- Operational Sensitivity: Risk levels are adjusted based on how hard the equipment is worked and how well it is maintained. This allows us to charge higher premiums for high-intensity zones like Bayesia compared to calmer systems.
- Stable Pricing: Our model maintains a consistent 5% profit margin and fixed protective buffers (multipliers) to ensure long-term solvency and pricing transparency.

Workers' Compensation

Metric	**Description**	**Rationale**
Claim frequency distribution	The claim frequency follows a Poisson distribution.	The mean and variance of the claim count are the same.
Claim severity distribution	The main part of the claim frequency follows a Gamma distribution.	The claim amount is right-tailed.
Claim timing	All claims are made at the end of the year.	To make the modelling process easier, cause the computing power is limited.
Exposure	The exposure for all policyholders is set to 1.	
Type of claim	All types of claims discussed in 2.4 are included in the data.	To make pricing and product design consistent, because there's no variable in the data that indicates the type of claim defined in this report.
Risk levels and exposures of new solar operation systems	The risk levels and exposures of new solar operation systems are set equal to solar systems with data	There's no information on how many insured will be covered in the new solar systems. As a conservative estimation, their risk levels and exposures are set equal to known solar systems.

8.6 Bibliography

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8.7 GitHub Link

For a deeper look into the data cleaning and modelling process, please access the GitHub link below:

https://github.com/seantran17/soaspacecase/